Report: SS Computer Project

Intro.

Basic Circuit characteristics
- speeds.
- power
- status - timescale

Machine Organization
- 2 or 3 possible layouts
- I/O considerations

calculations ~~of~~ for each
- basic circuit speeds
- line length considerations
- complexity of circuits on logic basis.

Evaluation

Recommendations

T = no. of Transistors

a = no. Trs per logic block

L = no. logic blocks v_b = volume of logic block

$$La = T$$

W_b = power generated per logic block (power density)

θ = Temperature C = heat capacity of body

[illegible]

$$C\frac{d\theta}{dt} = w - k\theta$$

power   conduction [illegible]

$Q = C\theta$ total heat

$W_b \cdot L = W$ total power = (Vol)(w_l power density)

$$Vol = L \cdot v_b \cdot f$$

f = factor = $\frac{\text{actual vol}}{\text{vol of logic blocks total}}$

(measure of packing)

$$L \cdot W_b = L v_b \cdot f \cdot w_l$$

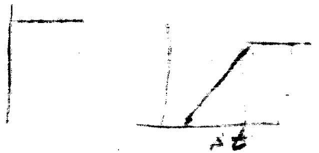
$$\frac{W_b}{v_b} = f w_l$$

$W_b = v_b w_b$ per block

$$w_b = f \cdot w_l$$

power density of block itself   power density of system

$$\frac{\partial Q}{\partial t} = w = EI = \frac{E^2}{R}$$



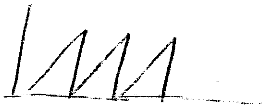
$E = bt$ $\left(\frac{dE}{dt}\right)$

~~$P = tE_0$~~

total heat
1 pulse

$$Q = \int_0^{\Delta t} w\,dt = \frac{\int (E_0 t)^2 dt}{R} = \frac{b t^3}{R}\Big|_0^{\Delta t} = \frac{b}{3R}(\Delta t)^3$$

$$E = Ri + L\frac{di}{dt}$$



$$Q = \frac{b}{3R}(\Delta t)^3$$

$$\frac{\partial Q}{\partial t} = \frac{E^2}{R} = \frac{E^2(t)}{R}$$

heat per unit time $\frac{Q}{\Delta t} = \frac{b}{3R}(\Delta t)^2$

$E_0 e^{\beta t} = R i_0 e^{\beta t} + L \beta i_0 e^{\beta t}$ $\quad i = i_0 e^{\beta t}$

$\frac{E_0}{R + L\beta} = i_0$

$E = Ri + L \cdot \theta i = (R + L\cdot\theta) i$

$\frac{E}{R + L\theta} = i$

$\frac{E/L}{R/L + \theta} = i$

look up d.e.

$\frac{L}{2\tau}$

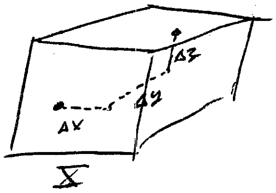
L = no. logical elements

b = average branching per element

f = frequency (cycle rate average)

h = quantity of heat released per cycle for one element

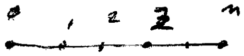
random distance between elements (assuming rectilinear wiring)



$l = \Delta x + \Delta y + \Delta z$ where Δx = average x distance between any two elements

$l = X$ (if cube)

$\Delta x = \frac{1}{3}X$ (see below)



$$\overline{\Delta x} = \frac{\sum_{j=0}^{n}\left(\sum_{i=0}^{n}|x_j - x_i|\right)}{(n+ \ )^2 \Delta x^2 = (\text{total dist})^2}$$

example $n=1$

	$j=0$	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$
$i=$	0	1	2	3	4	5
	1	0	1	2	3	4
	2	1	0	1	2	3
	3	2	1	0	1	2
	4	3	2	1	0	1
	5	4	3	2	1	0
	$\frac{15}{6}$	$\frac{11}{6}$	$\frac{9}{6}$	$\frac{9}{6}$	$\frac{11}{6}$	$\frac{15}{6}$

$\frac{\Sigma\Sigma}{6^2} = \frac{70}{36}$ = ~~1.94~~ 2.8 ? $\frac{1.94}{5} = 0.39$ (total)

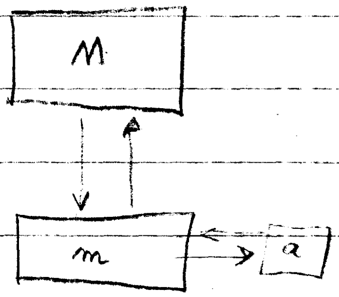
~~$A = \int_0^{X} \frac{(x-x_0)\,dx}{X} = \frac{1}{X}\left(\frac{1}{2}x^2 - x_0 x\right)\Big|_0^{X} = \frac{1}{2}X_0 - x_0$~~

~~$\overline{\Delta x} = \frac{\int_0^{X} A\,dx_0}{X} = \frac{\left(\frac{1}{2}X x_0 - \frac{1}{2}x_0^2\right)}{X}\Big|_{x_0=0}^{X} = \frac{1}{2}X - \frac{1}{2}X_0$~~

$$A = \frac{1}{X}\left[\int_0^{x_0}(x_0 - x)\,dx + \int_{x_0}^{X}(x - x_0)\,dx\right] = x\left(x_0 - \frac{1}{2}x\right)\Big|_0^{x_0} + x\left(\frac{1}{2}x - x_0\right)\Big|_{x_0}^{X} = \left[x_0\left(x_0 - \frac{1}{2}x_0\right) + X\left(\frac{1}{2}X - x_0\right) - x_0\left(\frac{1}{2}x_0 - x_0\right)\right]$$

$$= \left[\frac{1}{2}x_0^2 + X\left(\frac{1}{2}X - x_0\right) + \frac{1}{2}x_0^2\right]$$

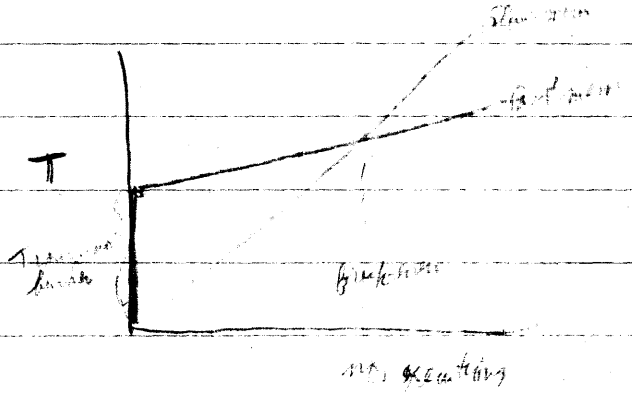
$$\Delta X = \frac{1}{(X)^2}\int_0^{X}\left(X\left(\frac{1}{2}X - x_0\right) + x_0^2\right)dx_0 = \frac{1}{X^2}\left(\frac{1}{3}X^3\right) = \frac{1}{3}X$$



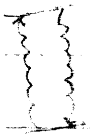
e_i = execution time for i^{th} instr $= a \cdot C_m$

$R = \frac{C_M}{C_m}$ = ~~ratio of fetch times (cycle time for 1 mem)~~
for the time each instr is used

$$T = N_{instr} \times C_M + e_i \cdot N \cdot \boxed{f} = N(C_M + a C_m f) = N C_m (R + af)$$



utilize



multiply
memory

Speed = Problems done/unit time = A·(Problem/time) = A·B(Ops/unit time) = A·B·C([illegible])

= A·B·$\frac{C \cdot E}{D}$(logical [illegible]/time) no. of [illegible] cycle

no. clocks/op